

WORKING

The working of the SafeSpin – Anti Suicide Fan system is based on detecting abnormal downward force applied to the ceiling fan rod and triggering a series of safety responses through an embedded electronic system. The system integrates a mechanical sensing arrangement with a microcontroller based monitoring unit to ensure early detection and rapid notification during emergency situations. Under normal operating conditions, the ceiling fan rotates normally and the mechanical components remain in their default positions. The clamp attached to the fan rod is positioned slightly above the roller limit switch with a small air gap of approximately 1–2 mm. This air gap is intentionally maintained to prevent the limit switch from being triggered due to minor vibrations produced during regular fan operation. The fan rod also contains a spring mechanism that allows slight axial movement when an external downward force is applied.

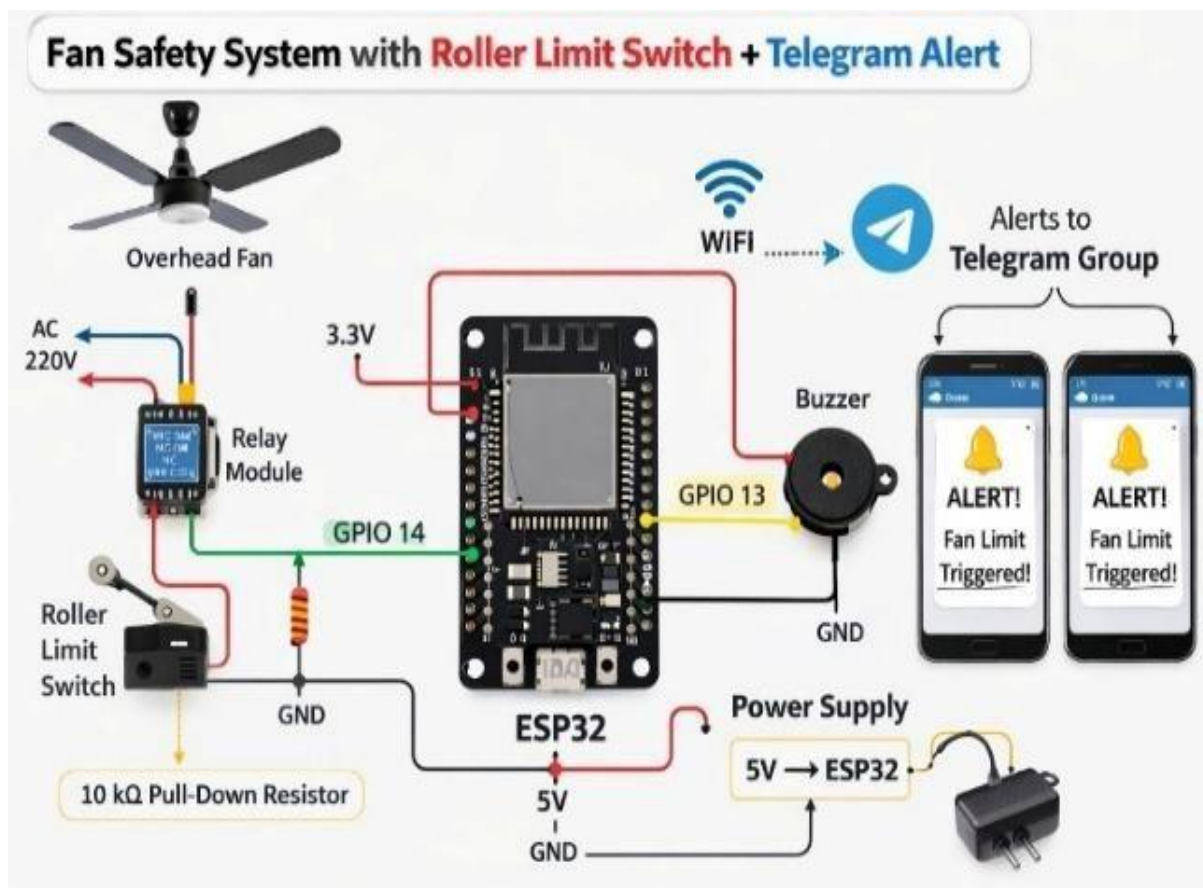
During normal usage, the spring remains in its stable position and no displacement occurs. However, when a significant downward force is applied to the fan rod, the spring mechanism allows the rod to move slightly downward. As the rod moves downward, the clamp attached to the rod also moves downward. When the downward displacement reaches a certain level, the clamp presses the roller limit switch mounted on the shelf bracket. Once the limit switch is pressed, it generates an electrical signal which is transmitted to the ESP32 microcontroller. The ESP32 continuously monitors the state of the limit switch, and when the switch is triggered it immediately recognizes the condition as an abnormal load event. After detecting the trigger signal, the ESP32 performs several programmed safety actions.

The first action is to activate the relay module connected to the ceiling fan power supply. The relay acts as an electrically operated switch that disconnects the power supply to the fan. This stops the fan operation and prevents any further mechanical stress on the system. At the same time, the ESP32 activates a buzzer that produces a loud audible alert. The purpose of the buzzer is to attract the attention of nearby individuals such as roommates, hostel wardens, or family members so that they can quickly respond to the situation.

In addition to the local alarm, the ESP32 uses its built-in Wi-Fi capability to send an emergency notification message through the Telegram messaging platform. The message is delivered instantly to predefined contacts such as hostel wardens, security staff, or caregivers. This ensures that responsible individuals are informed immediately even if they are not physically present near the location. The entire system operates automatically and continuously

monitors the condition of the fan rod without interfering with the normal functioning of the ceiling fan. The combination of mechanical sensing, embedded control, and wireless communication provides a reliable safety mechanism capable of detecting abnormal situations quickly.

By integrating a mechanical detection mechanism with an IoT-based alert system, the SafeSpin – Anti Suicide Fan system provides a practical and cost-effective approach to improving safety in environments such as hostels, hospitals, and residential buildings. The system ensures early detection of dangerous conditions and enables rapid response, thereby increasing the chances of preventing fatal incidents.



FLOWCHART

